Non-Conventional Logic Exploration for High-Speed Ladner–Fischer Prefix Adders

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ABSTRACT

The need for high-speed and low-power arithmetic circuits is increasing in modern VLSI systems. This work presents the design of a Ladner–Fischer prefix adder using non-conventional logic techniques to improve performance. The proposed approach focuses on reducing delay, power consumption, and circuit area. Compared to traditional designs, the proposed adder uses fewer components and reduces switching activity, which helps in achieving faster operation and lower energy usage. The structure of the adder is optimized to improve speed and make the design more efficient. The results show that the proposed design provides better performance in terms of speed and power when compared to conventional methods. It also offers good scalability, making it suitable for use in high-speed and low-power digital applications.

Keywords: Ladner–Fischer Adder, Prefix Adder, Non-Conventional Logic, High-Speed Design, Low Power Design, VLSI Circuits, Delay Reduction, Area Optimization, Efficient Circuits

INTRODUCTION

Arithmetic circuits form the backbone of digital systems, where addition is one of the most fundamental and frequently executed operations. The performance of these systems is largely influenced by the efficiency of the adder architecture used. As the need for high-speed and low-power computation continues to increase, designing optimized adders has become a critical research focus.

Parallel prefix adders are widely adopted due to their capability to minimize carry propagation delay. Among them, the Ladner–Fischer prefix adder is known for providing an effective balance between computational depth and structural complexity. Its hierarchical carry generation mechanism enables faster operation compared to traditional adder designs.

Conventional implementations of prefix adders primarily rely on standard CMOS logic, which, although robust, often leads to increased power consumption and larger circuit area. To address these drawbacks, alternative design techniques such as Gate Diffusion Input (GDI) have been introduced. GDI-based designs significantly reduce transistor count and power usage while maintaining acceptable performance, making them a preferred choice in many existing approaches.

However, despite these improvements, further optimization is still required to meet the demands of modern high-speed applications. This motivates the exploration of additional non-conventional logic styles that can enhance performance beyond the limitations of existing GDI-based implementations.

In this work, a detailed exploration of non-conventional logic techniques is carried out for the design of high-speed Ladner–Fischer prefix adders. The objective is to evaluate their impact on speed, power efficiency, and overall circuit performance, thereby identifying more effective design strategies for advanced digital systems.

MATERIALS AND METHODS

Design Approach:

The proposed design adopts a structured and performance-driven approach for implementing a high-speed and low-power Ladner–Fischer parallel prefix adder using non-conventional logic techniques. The architecture is fundamentally based on parallel prefix computation, where in the carry generation process is transformed into a tree-based structure to minimize propagation delay. The Ladner–Fischer topology is specifically chosen due to its balanced trade-off between logic depth,fan-out and interconnection complexity, making it suitable for scalable VLSI implementations.

The design begins with the formulation of propagate and generate signals, which serve as the core elements for prefix computation. These signals are processed through multiple hierarchical stages, enabling concurrent evaluation of intermediate carries. Unlike linear carry propagation in conventional adders, this parallel evaluation significantly reduces the critical path length, thereby enhancing computational speed. The prefix network is carefully structured to avoid excessive fan-out while maintaining optimal node distribution across stages.

To further improve efficiency, the design replaces traditional static CMOS logic with advanced non-conventional logic styles, including Gate Diffusion Input (GDI) and Pass Transistor Logic (PTL). These techniques are strategically integrated into the prefix nodes to reduce transistor count and parasitic capacitance. As a result, the switching activity is minimized, leading to lower dynamic power consumption and faster signal transitions. The hybrid utilization of these logic styles ensures that the design achieves both compactness and high performance without compromising reliability.

Critical path optimization is a key focus of the design approach. The arrangement of prefix stages is refined to minimize logic levels, and transistor sizing is employed to balance drive strength across different nodes. Additionally, buffering strategies are incorporated where necessary to maintain signal integrity and prevent degradation due to capacitive loading. The interconnect structure is also optimized to reduce delay caused by wiring resistance and capacitance, which are significant factors in deep submicron technologies.

The design is further validated through a layout-aware methodology, ensuring that the physical implementation aligns with the intended performance objectives. Placement and routing considerations are incorporated early in the design process to reduce area overhead and interconnect complexity. Post-layout simulations are conducted to capture the impact of parasitic elements and verify the robustness of the design under realistic operating conditions.

Overall, the proposed design approach emphasizes a synergistic combination of efficient prefix architecture and optimized logic styles to achieve superior performance. The resulting adder demonstrates improved speed, reduced power consumption, and enhanced scalability, making it well-suited for high-performance digital systems and advanced VLSI applications.

Non-conventional parallel prefix adder design:

The design of parallel prefix adders using non-conventional logic techniques represents a significant advancement in modern VLSI systems aimed at achieving high-speed and energy-efficient computation. As technology scales into nanometer regimes, traditional static CMOS logic encounters critical challenges including increased leakage power, larger area overhead, and interconnect-induced delays. To address these limitations, non-conventional logic styles such as Gate Diffusion Input (GDI), Pass Transistor Logic (PTL), and transmission gate-based implementations have been extensively explored to enhance arithmetic circuit performance.

In a parallel prefix adder, the computation of carry signals through generate and propagate functions is inherently optimized using a tree-based structure, enabling logarithmic delay characteristics. By incorporating GDI-based designs, the transistor count required for implementing prefix operations is significantly reduced, leading to lower switching activity and improved power efficiency. Furthermore, PTL-based implementations minimize capacitive loading and eliminate redundant logic transitions, thereby accelerating signal propagation along the critical path. The integration of transmission gate logic ensures full voltage swing and mitigates threshold voltage loss, enhancing signal integrity across multiple stages of the prefix network.

Ladner-Fischer Prefix adder :

The Ladner–Fischer prefix adder is a widely recognized parallel prefix architecture that provides an efficient trade-off between computational speed, wiring complexity, and area utilization in high-performance VLSI systems. Unlike conventional ripple carry adders, which suffer from linear propagation delay, the Ladner–Fischer structure employs a logarithmic-depth prefix tree to compute carry signals in parallel, thereby significantly improving overall addition speed. The fundamental operation of this adder is based on the generation and propagation of carry signals using associative prefix operators, which enable efficient merging of intermediate results across multiple stages. The architecture is characterized by a flexible tree structure that reduces fan-out while maintaining moderate interconnect complexity, making it more area-efficient compared to highly parallel designs such as the Kogge–Stone adder.

In advanced implementations, the Ladner–Fischer prefix adder can be further optimized using non-conventional logic techniques, which enhance its performance in deep submicron technologies. By integrating logic styles such as Gate Diffusion Input and pass transistor logic, the design achieves reduced transistor count, lower switching activity, and improved power efficiency.

Existing Method :

In existing VLSI design methodologies, the implementation of parallel prefix adders, including the Ladner–Fischer architecture, relies on a combination of conventional and semi-optimized logic styles such as static CMOS, Pass Transistor Logic (PTL), Dual Pass Transistor Logic (DPTL), and Gate Diffusion Input (GDI) techniques.

CMOS Ladner Fischer Prefix Adder :

The Ladner–Fischer prefix adder implemented using static CMOS logic is a high-performance arithmetic structure designed to compute binary addition with reduced propagation delay through parallel carry generation. In this architecture, the addition process is divided into three fundamental stages: pre-processing, prefix computation, and post-processing. During the pre-processing stage, the input bits are used to generate propagate and generate signals, where the propagate signal indicates whether a carry will propagate through a bit position, and the generate signal determines whether a carry is produced at that stage. These signals are typically realized using CMOS-based XOR and AND gate structures, ensuring full voltage swing and reliable operation.

For the given inputs A = 1000 and B = 0111, the least significant bit corresponds to the rightmost position. The propagate signals are obtained by performing bitwise XOR between A and B, resulting in a propagate sequence that reflects whether each bit pair can pass an incoming carry. Simultaneously, the generate signals are computed using bitwise AND operations, indicating the positions where a carry is inherently generated. Once these initial signals are established, the Ladner–Fischer prefix network evaluates carry signals in a tree-like structure. Unlike linear adders, this prefix network computes carries in parallel by combining intermediate generate and propagate pairs using associative operations, significantly reducing the critical path delay.

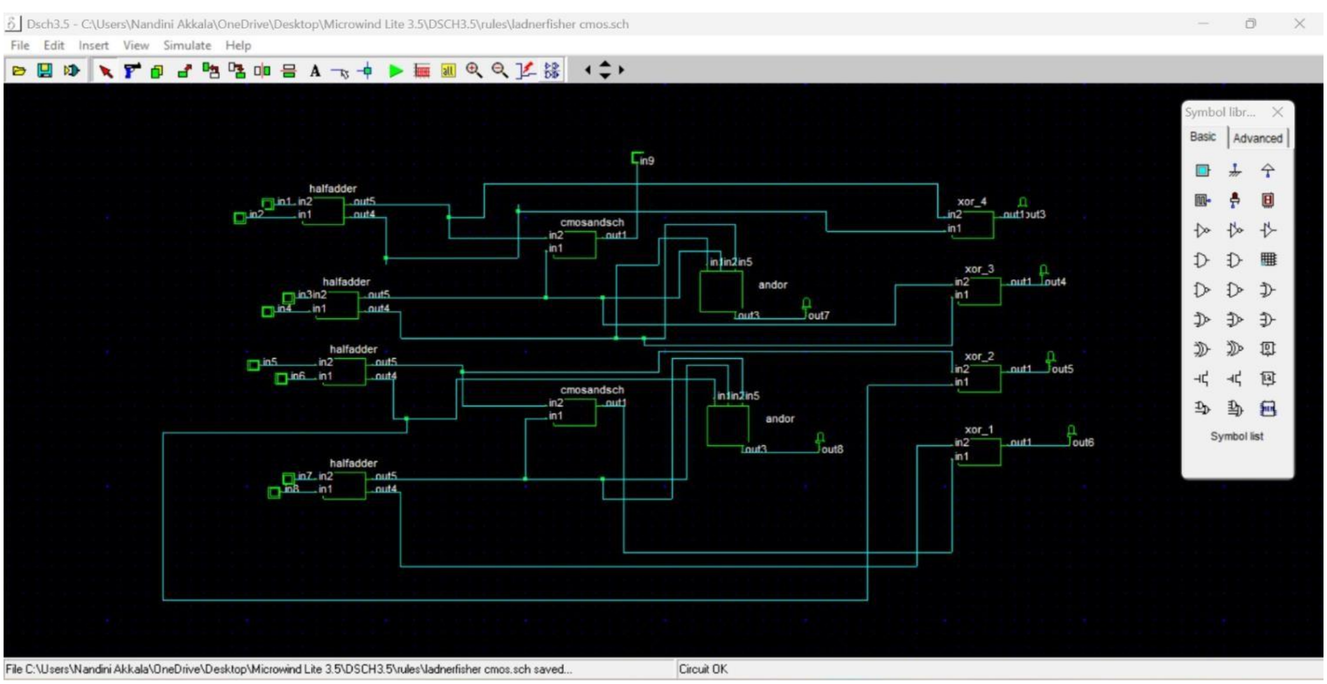


Fig.1: Ladner-Fischer Adder design using CMOS

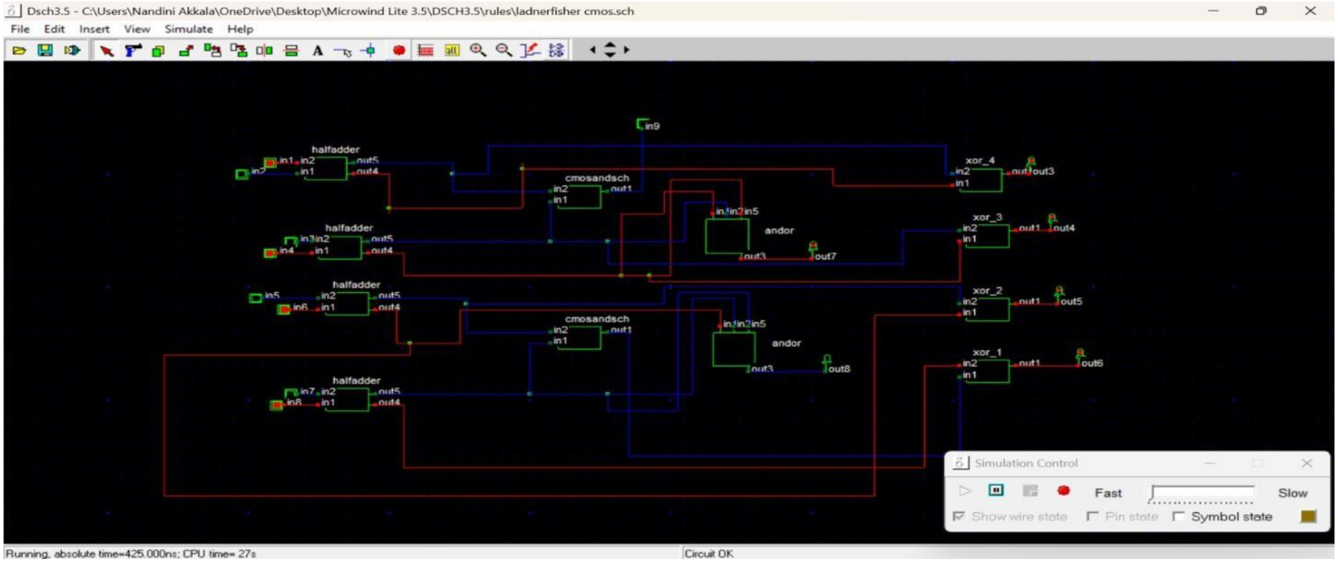


Fig.2: Execution of Ladner-Fischer Adder desing using CMOS

Fig.3: Layout Simulation of Ladner-Fischer Adder using CMOS

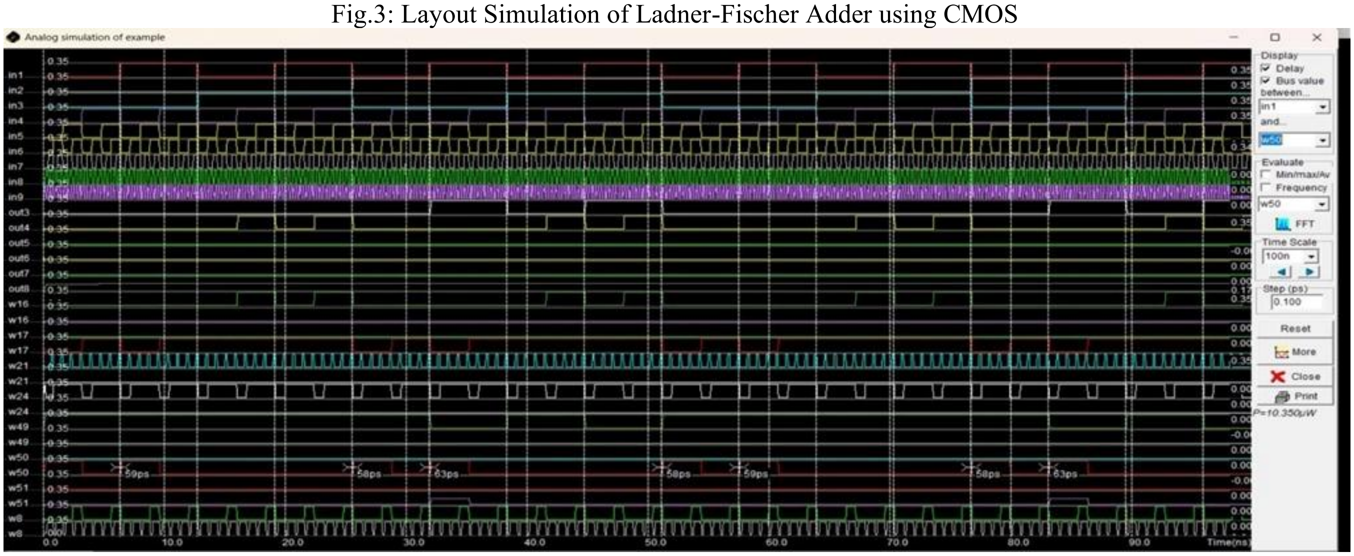


Fig.4: Power Results of Ladner-Fischer Adder using CMOS

PTL Ladner Fischer Prefix Adder :

The Ladner–Fischer prefix adder implemented using Pass Transistor Logic (PTL) represents an energy-efficient and high-speed alternative to conventional CMOS-based designs, particularly suitable for deep submicron VLSI systems. In PTL-based architectures, logic functions are realized by allowing transistors to directly pass input signals between nodes, thereby eliminating the need for complementary pull-up and pull-down networks. This significantly reduces transistor count, lowers capacitive loading, and enhances switching speed, which is critical in parallel prefix structures where multiple carry computations occur simultaneously.

For the given input values, the carry signals are propagated through the prefix network without significant delay due to the continuous propagate condition. As a result, the carry generated at the least significant stage effectively influences higher-order bits. In the post-processing stage, the final sum is computed using PTL-based XOR structures that combine propagate signals with the corresponding carry inputs. The resulting sum for A = 1000 and B = 0111 is 1111, demonstrating correct arithmetic functionality with efficient carry propagation.

The prefix adder operates using generate (G) and propagate (P) signals:

Generate:

Gi = Ai . Bi

Propagate:

Pi = Ai XOR Bi

These signals are combined using the prefix operator:

(Gk,Pk).(Gj,Pj)=(Gk+PK . Gj,;Pk . PJ)

This operation enables fast carry computation across multiple bits

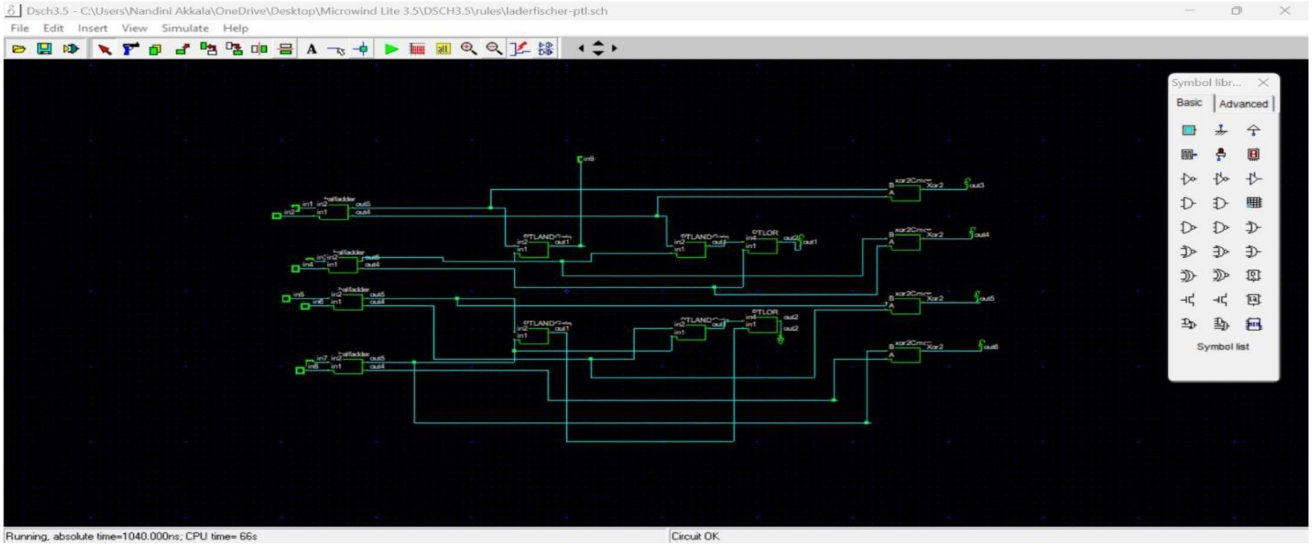


Fig.5: Ladner-Fischer Adder design Using PTL

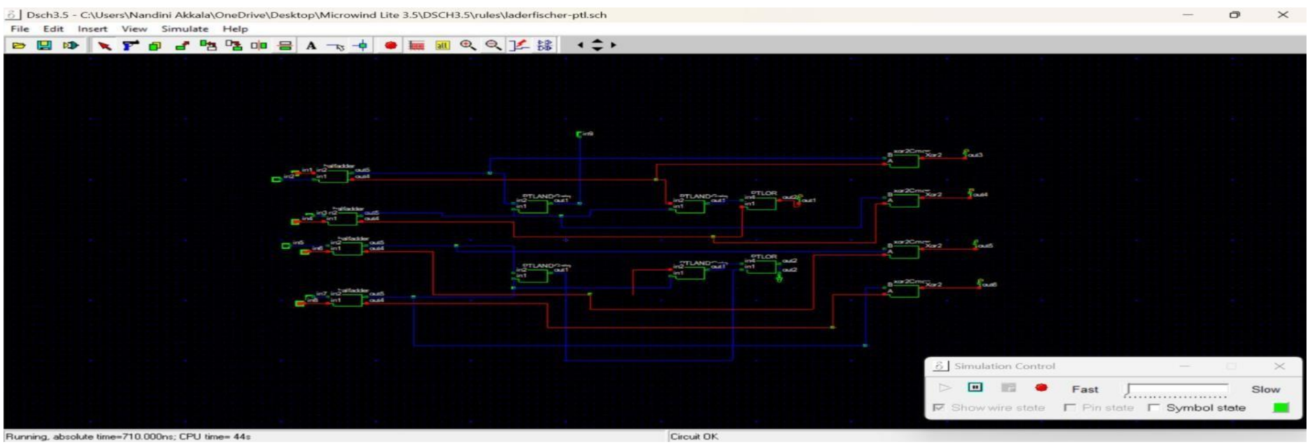


Fig.6: Execution of Ladner-Fischer Adder design using PTL

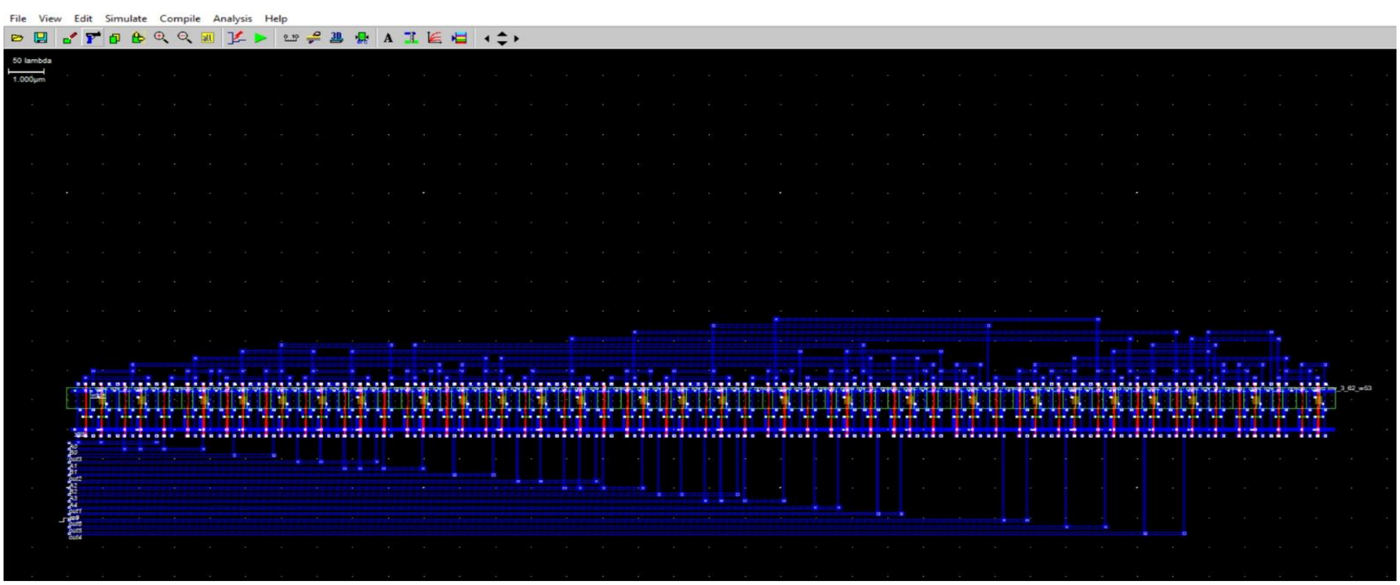


Fig.7: Layout Simulation of Ladner-Fischer Adder using PTL

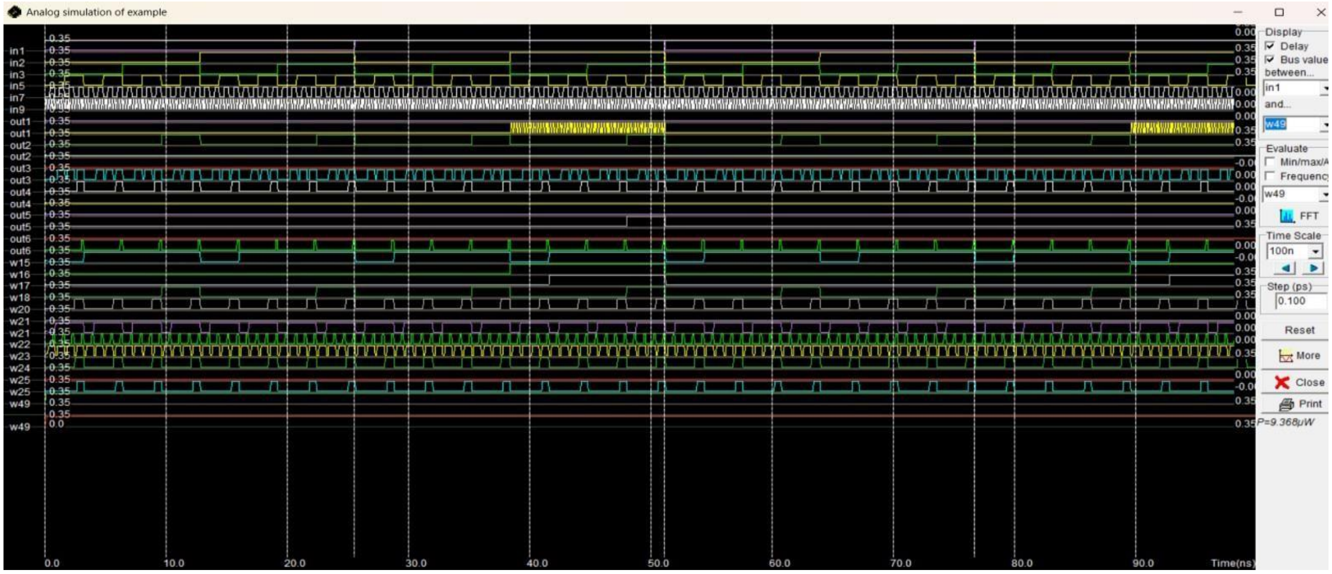


Fig.8: Power Results of Ladner-Fischer Adder using PTL

DPTL Ladner Fischer Prefix Adder :

The Ladner–Fischer prefix adder implemented using Dual Pass Transistor Logic (DPTL) represents a refined approach to high-speed arithmetic design by combining the advantages of pass transistor operation with improved signal integrity. Unlike conventional Pass Transistor Logic, DPTL utilizes both NMOS and PMOS transistors in parallel paths to transmit logic signals, thereby ensuring full voltage swing and reducing the threshold voltage degradation commonly associated with single-type pass transistor designs. This characteristic makes DPTL particularly suitable for multi-stage prefix structures such as the Ladner–Fischer adder, where signals must propagate reliably across several logic levels.

For the given input combination, the propagate condition allows the carry generated at the least significant stage to travel through successive prefix levels without obstruction. As a result, the carry signal effectively reaches higher-order bits with minimal delay. In the post-processing stage, the final sum outputs are computed by combining the propagate signals with the corresponding carry inputs using DPTL-based XOR structures. The resulting sum for A = 1000 and B = 0111 is 1111, with no overflow carry, confirming the correctness of the computation.

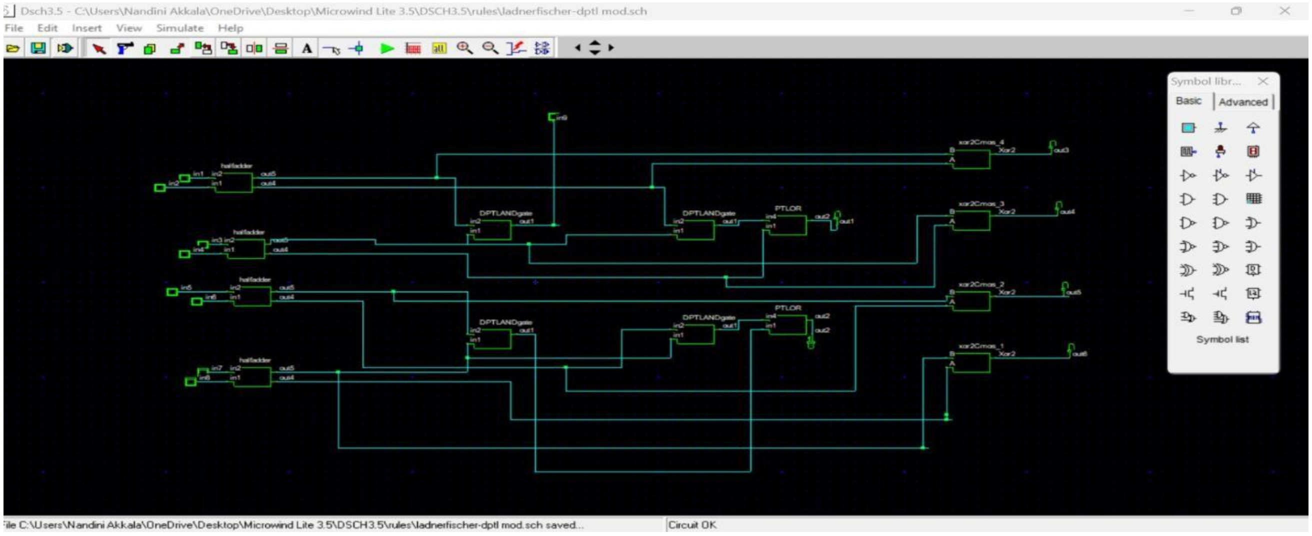


Fig.9: Ladner-Fischer Adder design using DPTL

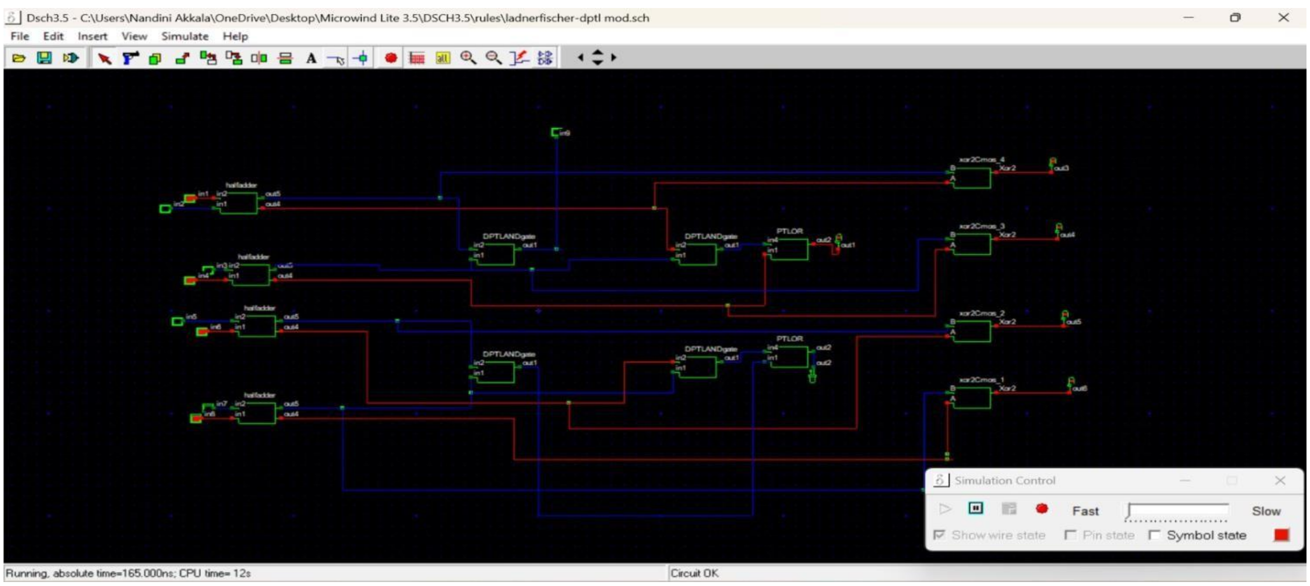


Fig.10: Execution of Ladner-Fischer Adder using DPTL

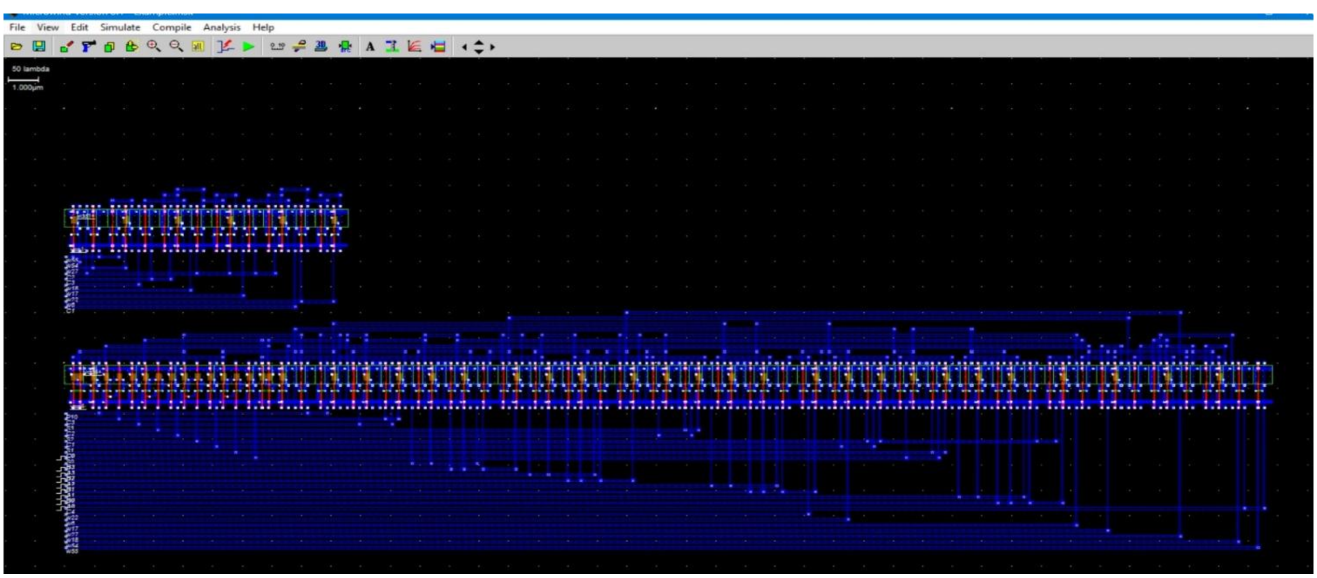


Fig.11: Layout Simulation of Ladner-Fischer Adder using DPTL

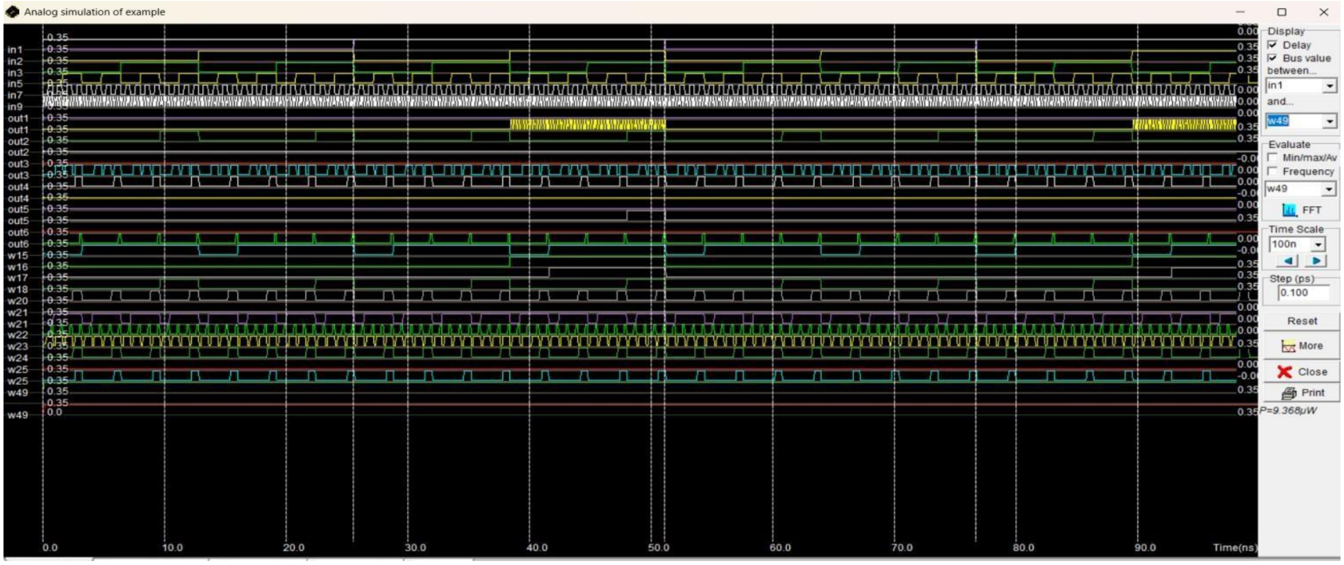


Fig.12: Power Results of Ladner-Fischer Adder using DPTL

GDI Ladner Fischer Prefix Adder :

The Ladner–Fischer prefix adder implemented using Gate Diffusion Input (GDI) logic represents a highly efficient approach for designing low-power and area-optimized arithmetic circuits in advanced VLSI systems. GDI logic differs fundamentally from conventional CMOS by allowing inputs to be applied not only at the gate terminals but also at the source and drain of transistors, enabling the realization of complex logic functions with a significantly reduced number of devices. This characteristic makes GDI particularly suitable for parallel prefix structures such as the Ladner–Fischer adder, where multiple generate and propagate operations must be computed efficiently across several stages.

For the given input combination, the continuous propagate condition allows the carry signal to pass through all stages of the prefix network without obstruction. As a result, the carry generated at the least significant stage influences all higher-order bits. In the post-processing stage, the final sum outputs are computed using GDI-based XOR structures that combine propagate signals with their corresponding carry inputs. The resulting sum for A = 1000 and B = 0111 is 1111, with no overflow carry, confirming correct arithmetic functionality.

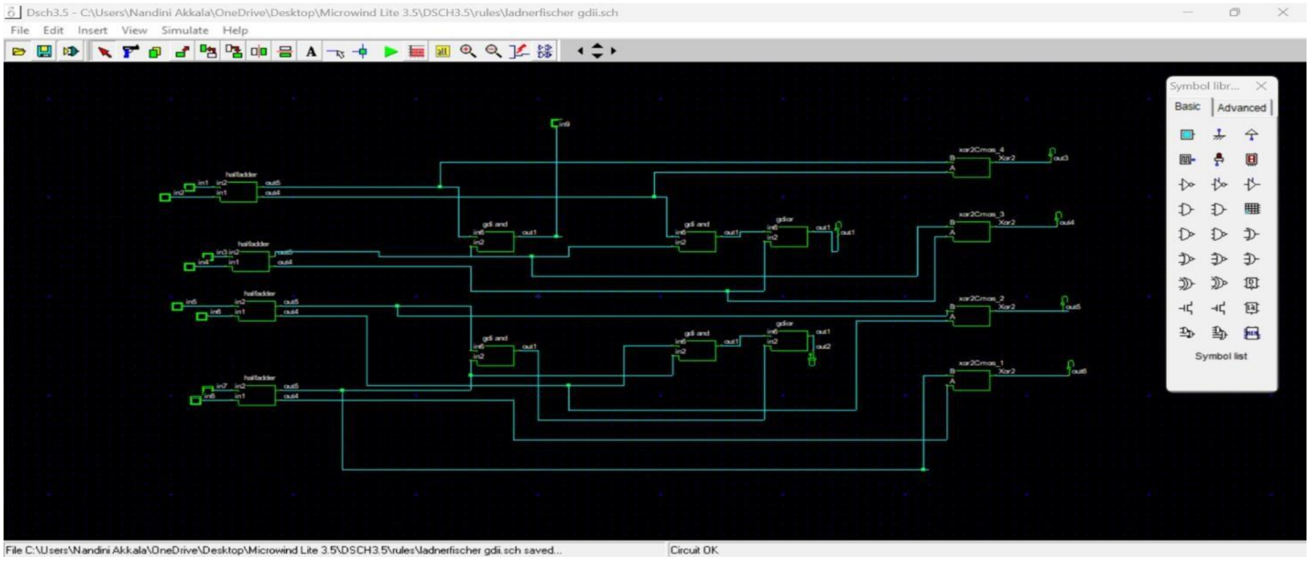


Fig.13: Ladner-Fischer Adder design using GDI

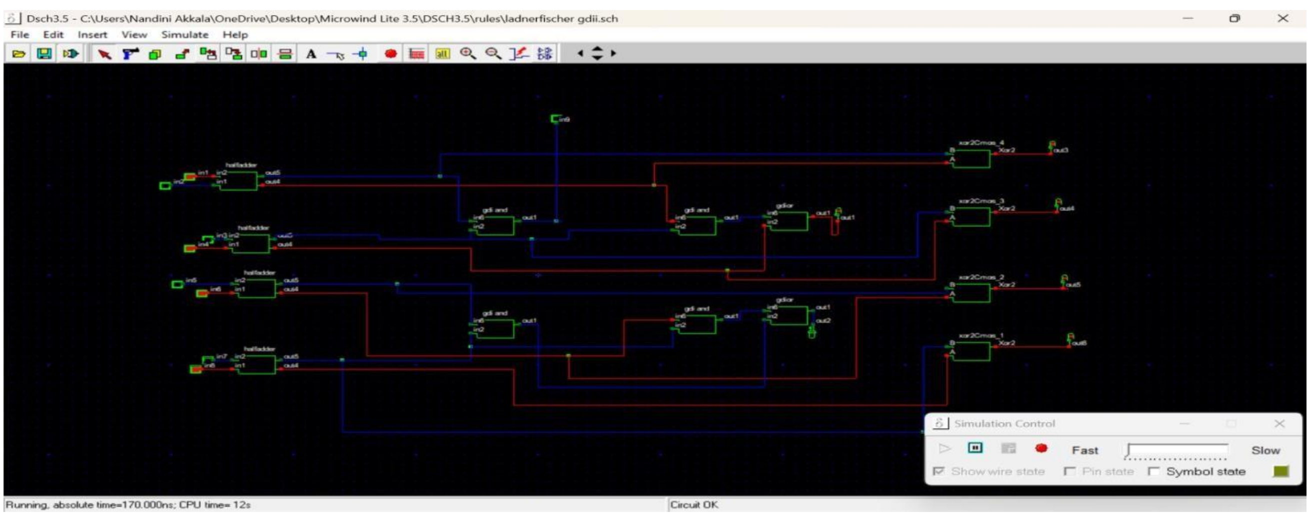


Fig.14: Execution of Ladner-Fischer Adder design using GDI

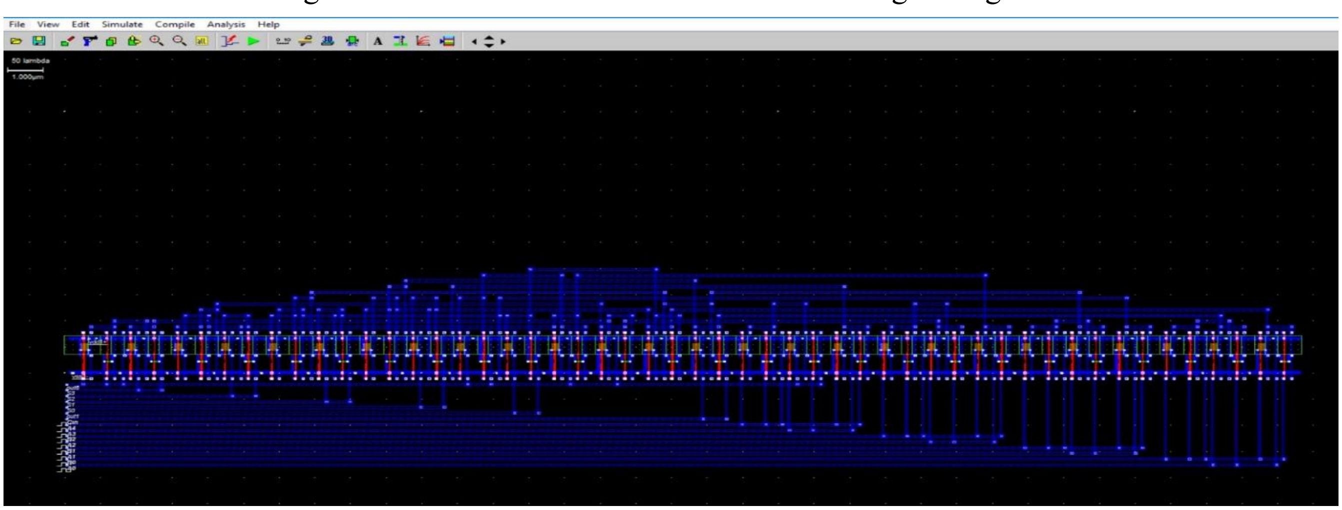


Fig.15: Layout Simulation of Ladner-Fischer Adder using GDI

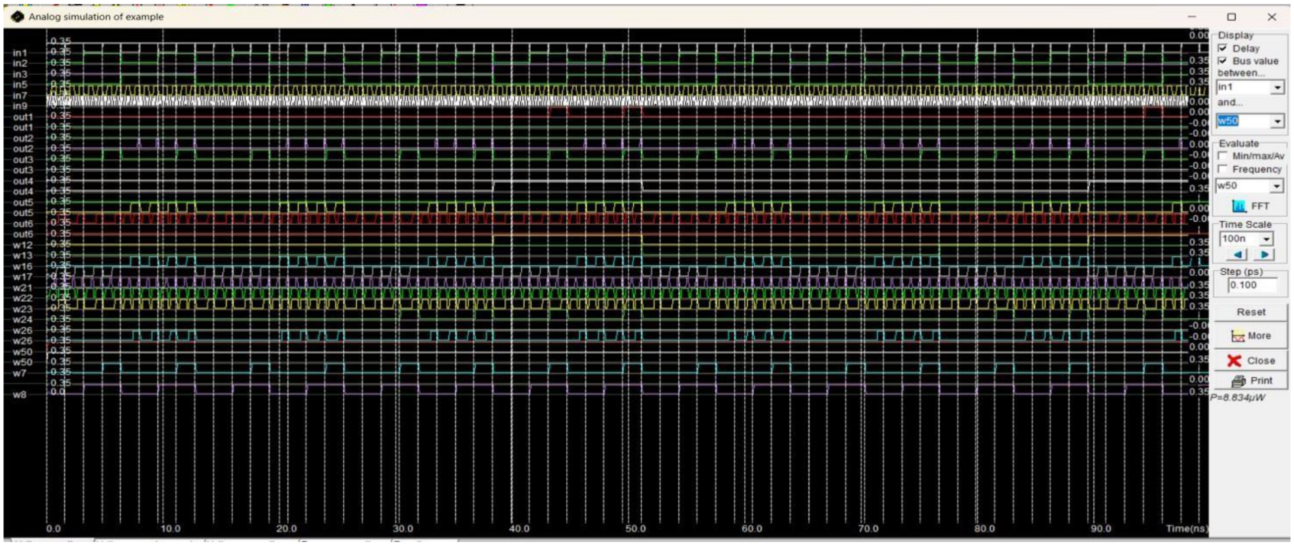


Fig.16: Power Results of Ladner-Fischer Adder using GDI

Proposed Method :

The proposed methodology introduces an enhanced design of the parallel prefix adder architecture by integrating Modified Gate Diffusion Input (MGDI) and Full Swing Gate Diffusion Input (FSGDI) logic techniques to achieve superior performance in advanced VLSI systems. As conventional logic styles encounter increasing limitations in nanometer-scale technologies, the adoption of optimized GDI variants provides an effective solution for reducing power consumption, minimizing area, and improving operational speed. The MGDI technique extends the basic GDI concept by refining input configurations and biasing conditions, thereby enabling the realization of complex logic functions with fewer transistors while maintaining improved signal reliability. This modification significantly reduces switching activity and internal node capacitance, which directly contributes to lower dynamic power dissipation and enhanced computational efficiency.

MGDI Ladner Fischer Prefix Adder :

The Ladner–Fischer prefix adder implemented using Modified Gate Diffusion Input (MGDI) logic represents an advanced and energy-efficient approach for high-speed arithmetic operations in modern VLSI systems. MGDI is an enhanced version of the conventional GDI technique, designed to overcome limitations related to threshold voltage loss and reduced output swing by optimizing input configurations and biasing conditions. This enables the realization of complex logic functions with fewer transistors while maintaining improved signal integrity, making MGDI particularly suitable for parallel prefix structures that require efficient carry propagation across multiple stages.

For the given input combination, the continuous propagate condition allows the carry signal to traverse all prefix levels efficiently without interruption. As a result, the carry generated at the least significant bit propagates through the network and contributes to the computation of higher-order carries. In the post-processing stage, the final sum outputs are obtained using MGDI-based XOR structures that combine propagate signals with their respective carry inputs. The resulting sum for A = 1000 and B = 0111 is 1111, with no overflow carry, confirming correct arithmetic operation.

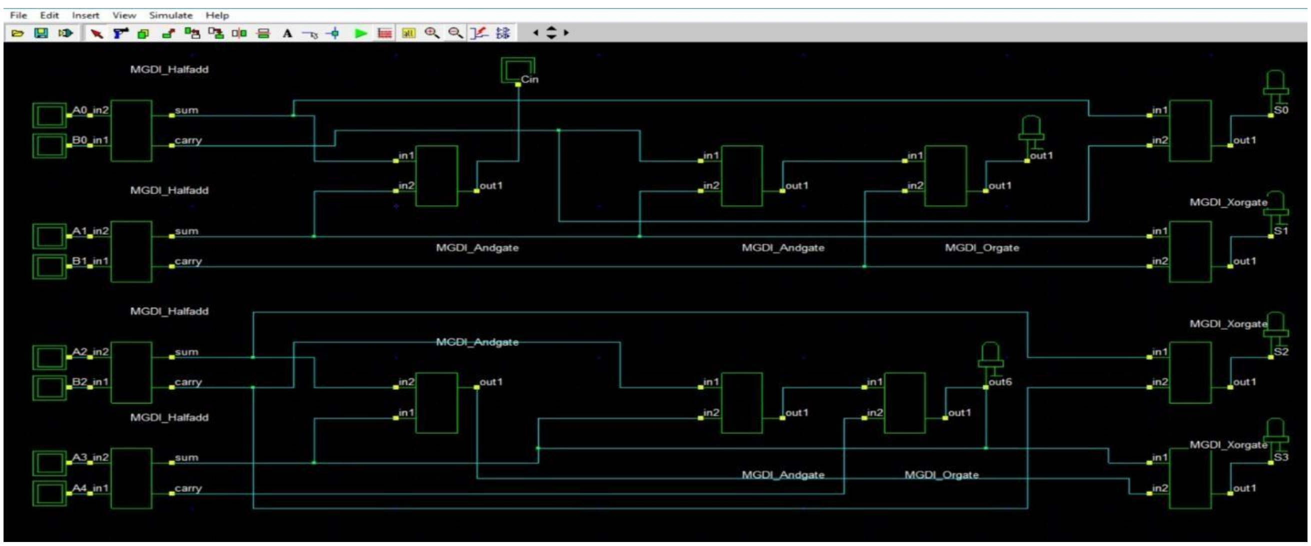


Fig.17: Ladner-Fischer Adder design using MGDI

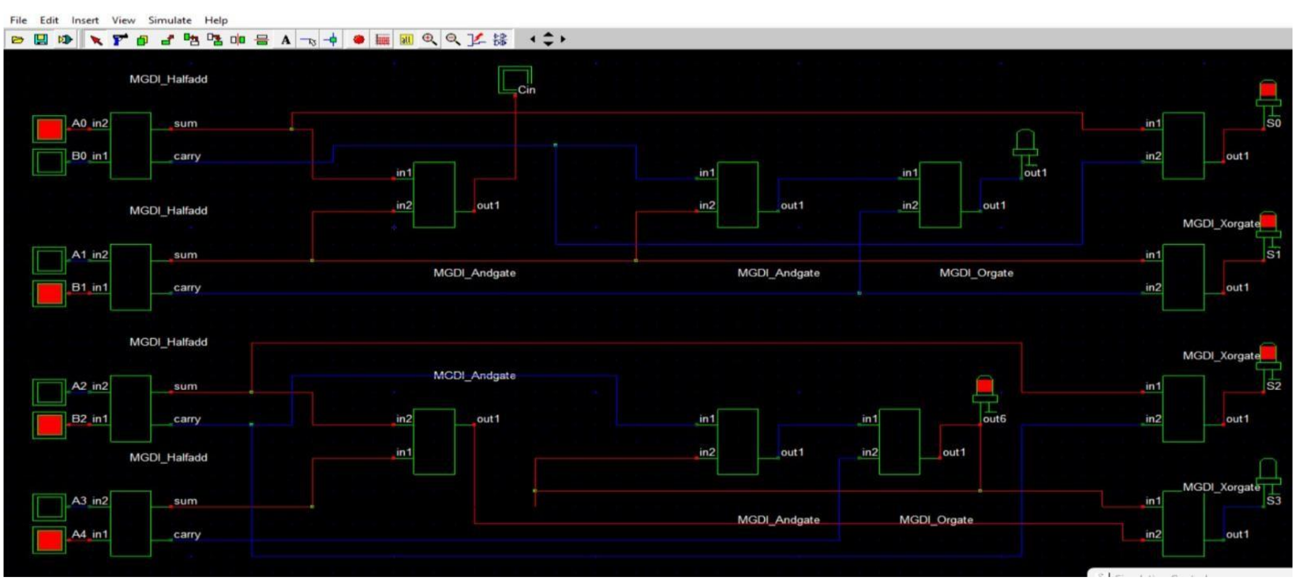


Fig.18: Execution of Ladner-Fischer Adder design using

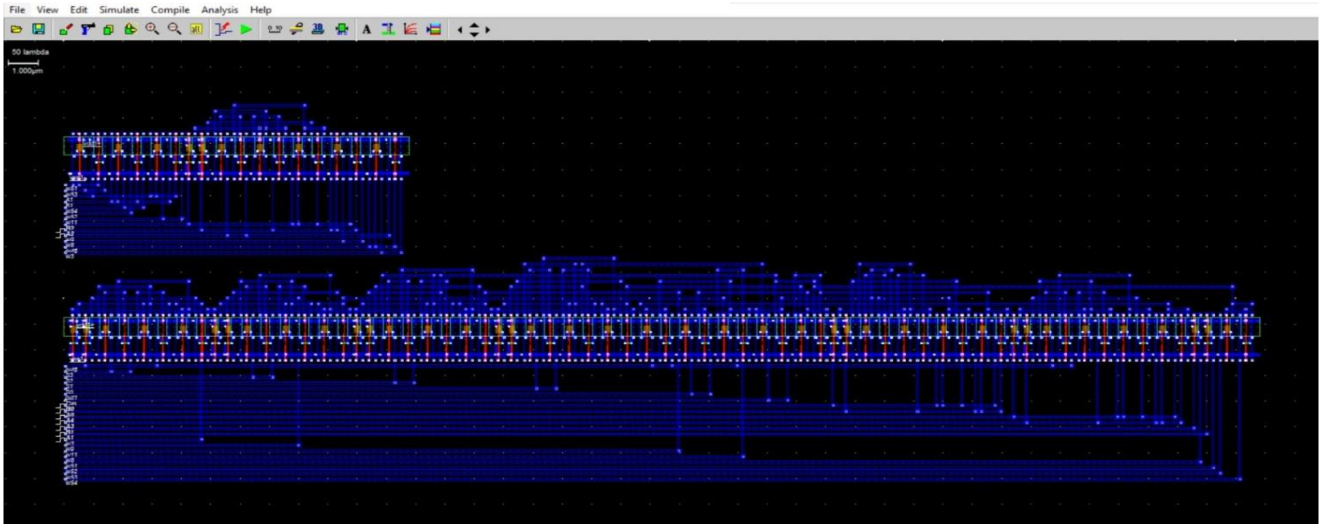


Fig.19: Layout Simulation of Ladner-Fischer Adder using MGDI

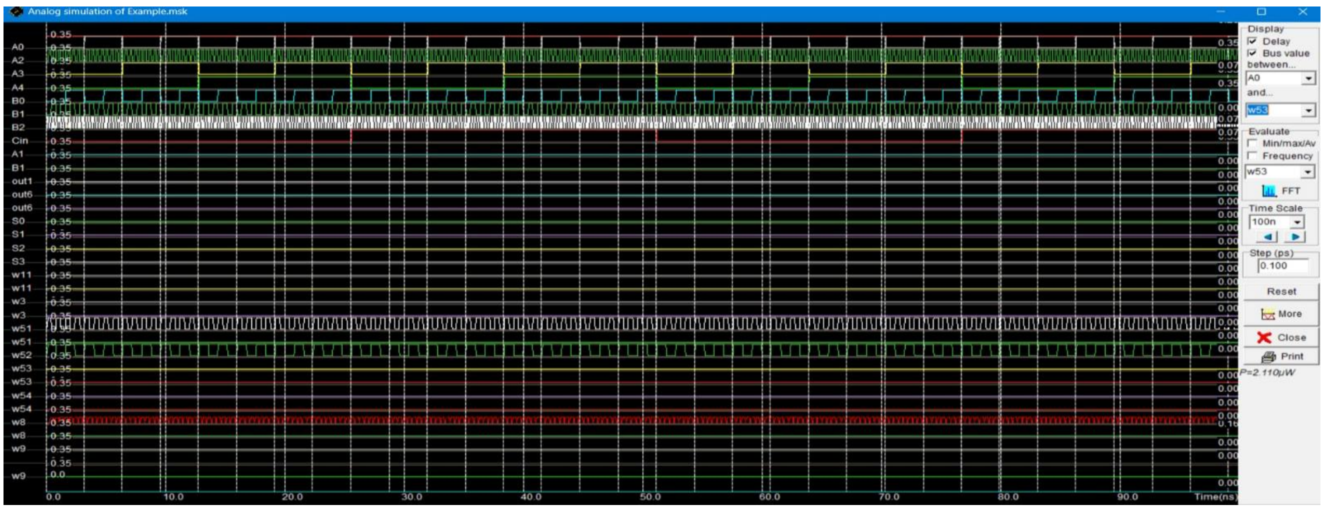


Fig.20: Power results of Ladner-Fischer Adder using MGDI

FSGDI Ladner Fischer Prefix Adder :

The Ladner–Fischer prefix adder implemented using Full Swing Gate Diffusion Input (FSGDI) logic provides an efficient solution for high-speed and low-power arithmetic operations in advanced VLSI systems. FSGDI enhances conventional GDI by ensuring full voltage swing, thereby eliminating signal degradation across multiple stages. In this architecture, addition is performed through pre-processing, prefix computation, and post-processing stages. For inputs A = 1000 and B = 0111, propagate signals are active at all bit positions, while generate signals remain in active. During prefix evaluation, carry signals are computed in parallel using the Ladner–Fischer tree, enabling reduced propagation delay. FSGDI-based logic ensures strong signal transmission with minimal transistor usage, reducing power consumption and improving speed. In the final stage, sum outputs are generated by combining propagate signals with carry inputs, resulting in 1111. This design achieves improved energy efficiency, reduced delay, and reliable operation, making it suitable for high-performance digital systems.

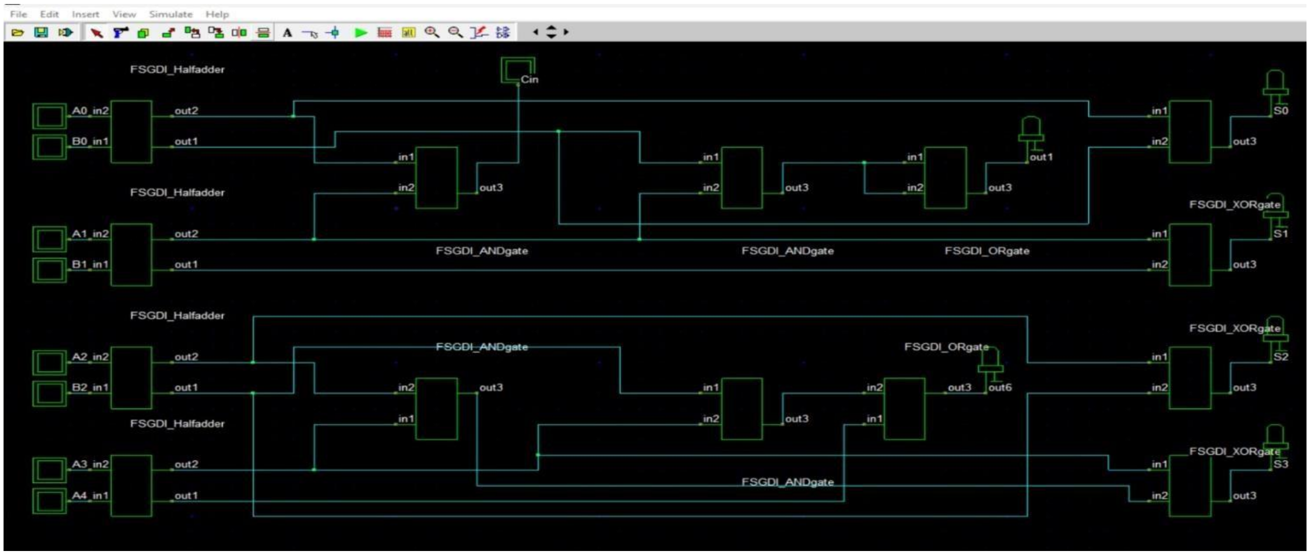


Fig.21: Ladner-Fischer Adder design using FSGDI

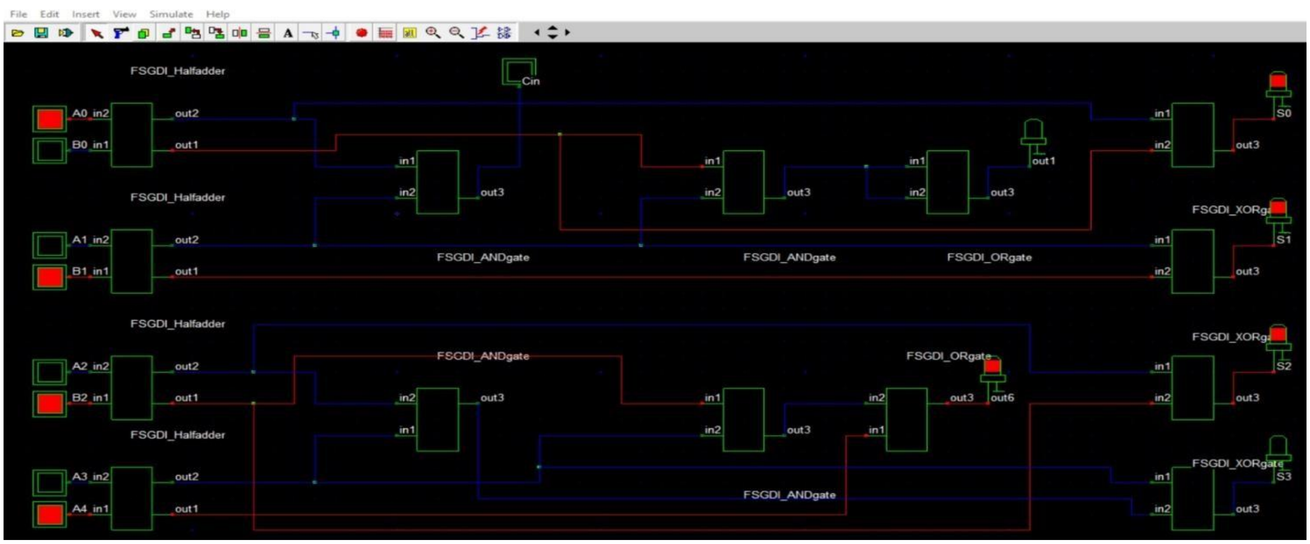


Fig.22: Execution of Ladner-Fischer Adder design using FSGDI

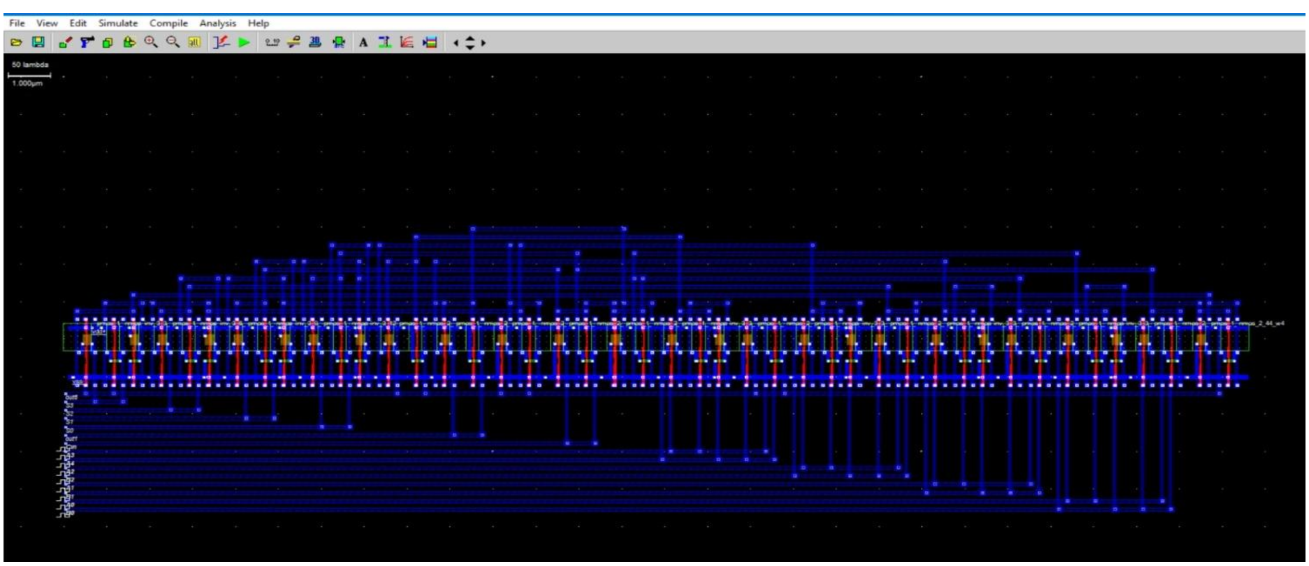


Fig.23: Layout Simulation of Ladner-Fischer Adder using FSGDI

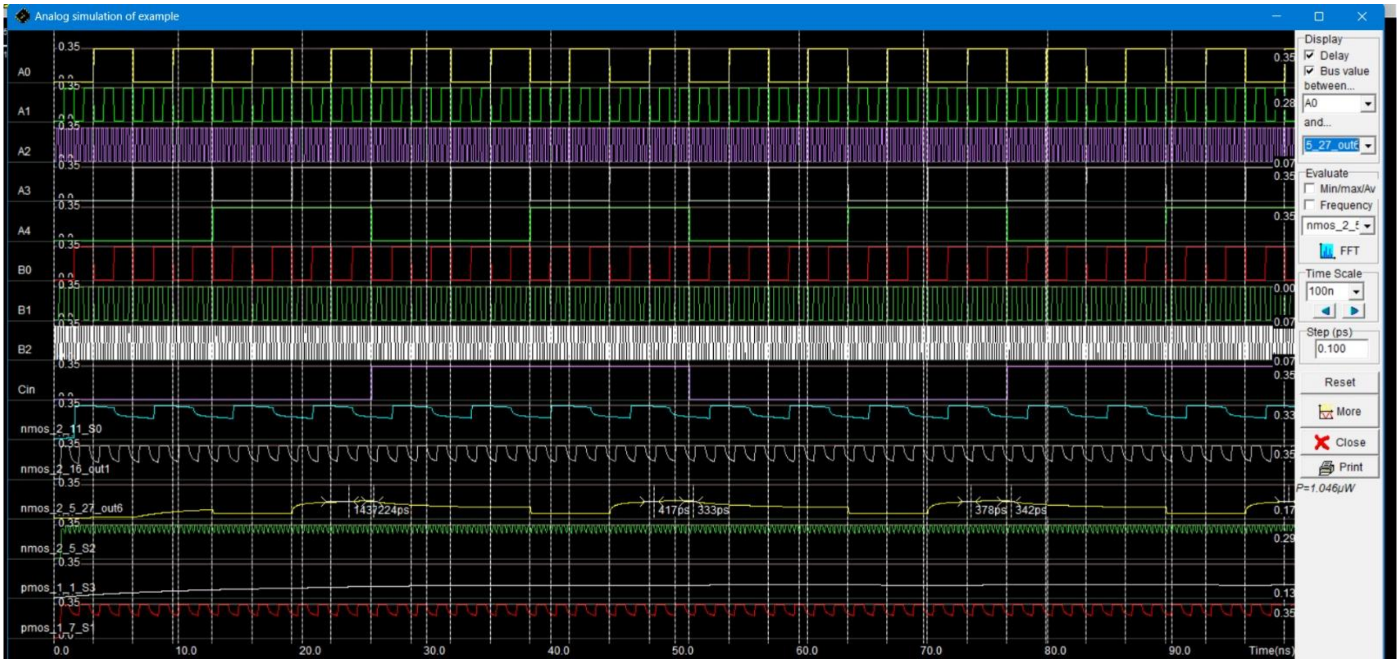


Fig.24: Power Results of Ladner-Fischer Adder using FSGDI

Transmission Gate Ladner Fischer Prefix Adder :

The Ladner–Fischer parallel prefix adder was previously implemented using Full-Swing Gate Diffusion Input (FSGDI) logic, which provided benefits in terms of reduced transistor count and compact layout but exhibited limitations in power efficiency and delay. To overcome these drawbacks, the design was extended using Transmission Gate (TG) technology, which ensures full voltage swing, eliminates threshold voltage drop issues, and reduces leakage currents. Simulation results confirmed that the TG-based implementation achieves lower dynamic power consumption and faster propagation compared to the FSGDI design, though at the cost of a slightly higher transistor count. Overall, the Transmission Gate extension enhances the performance of the Ladner–Fischer adder by delivering improved speed and energy efficiency, making it a more suitable choice for high-performance, low-power arithmetic applications.

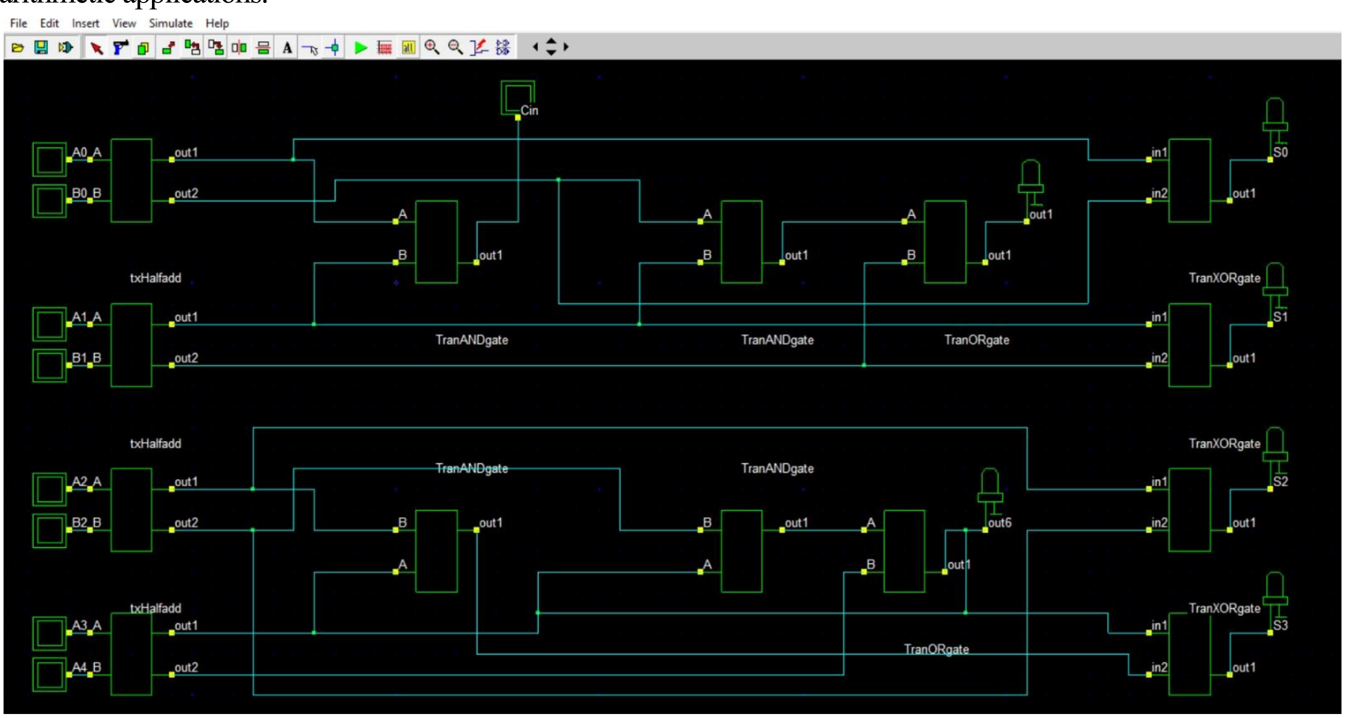


Fig.25: Ladner-Fischer Adder design using Transmission gate

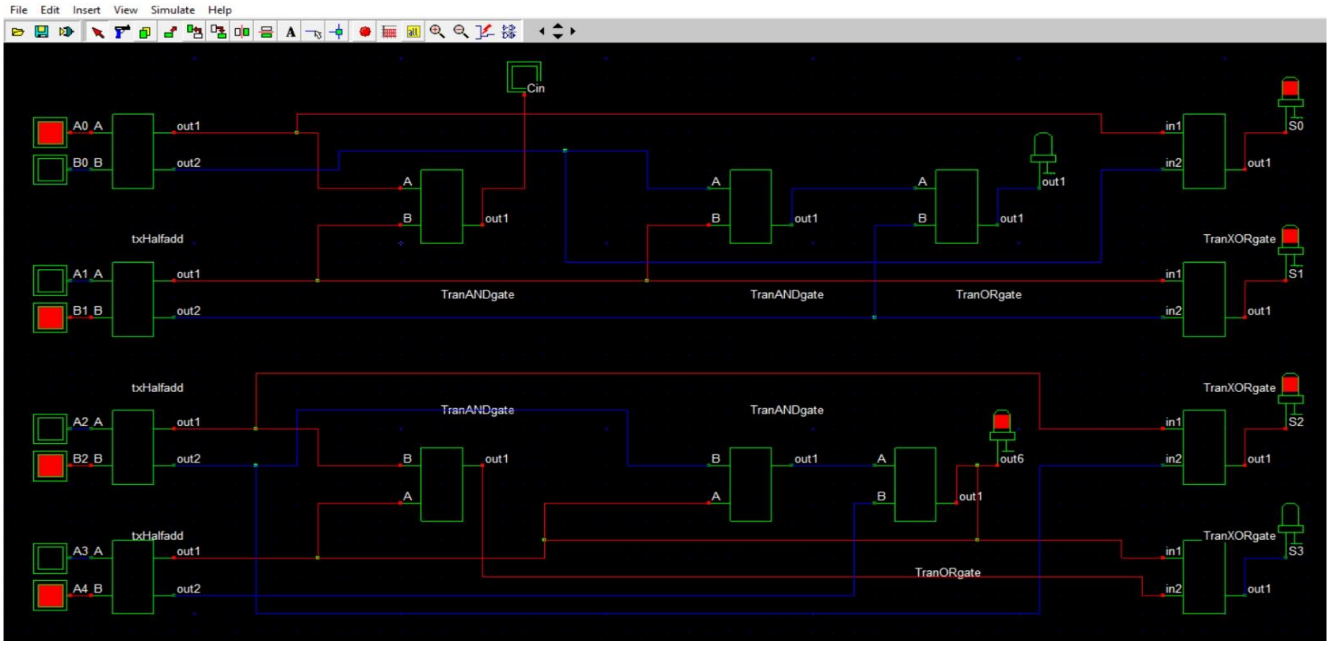


Fig.26: Execution of Ladner-Fischer Adder design using Transmission gate

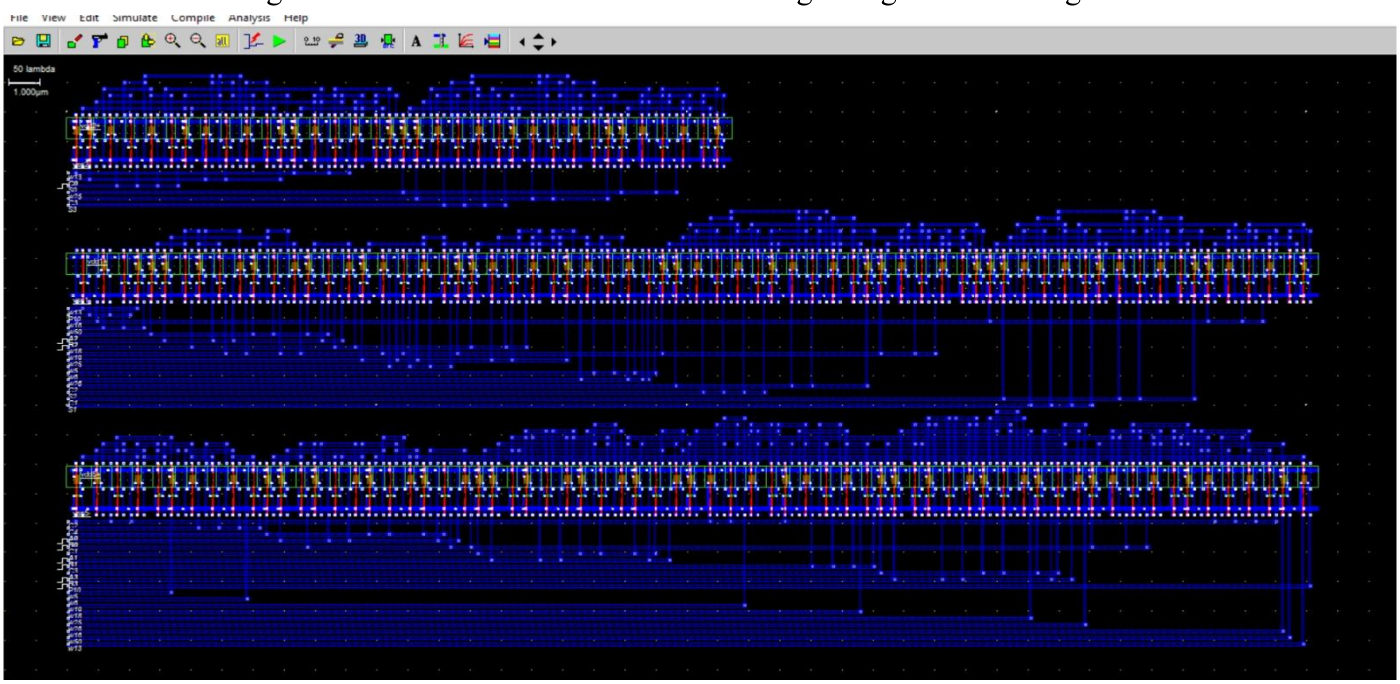
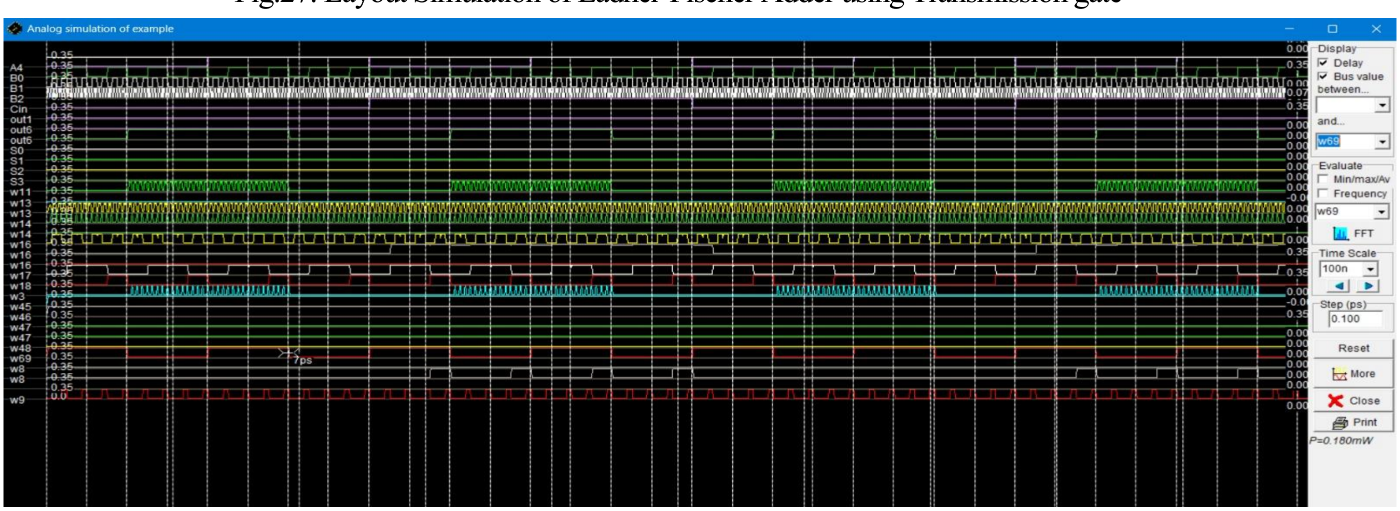


Fig.27: Layout Simulation of Ladner-Fischer Adder using Transmission gate



Fig,28: Power Results of Ladner-Fischer Adder using Transmission gate

FinFET Ladner Fischer Prefix Adder :

Building upon the Transmission Gate design, the Ladner–Fischer adder was further advanced using FinFET technology, which introduces a three-dimensional gate structure to achieve superior electrostatic control over the channel. Unlike planar TG circuits, FinFETs effectively suppress short-channel effects, minimize leakage currents, and deliver faster switching behavior. When integrated into the adder architecture, FinFET-based logic not only reduces both static and dynamic power consumption but also enhances propagation speed, making the design highly efficient for modern low-power, high-performance applications. Although the fabrication of FinFETs involves greater complexity compared to conventional CMOS or TG implementations, the resulting improvements in energy efficiency, scalability, and robustness establish FinFETs as a compelling solution for next-generation arithmetic units. This progression from FSGDI to TG and finally to FinFET highlights a clear trajectory toward achieving optimized adder designs with balanced trade-offs in speed, power, and reliability.

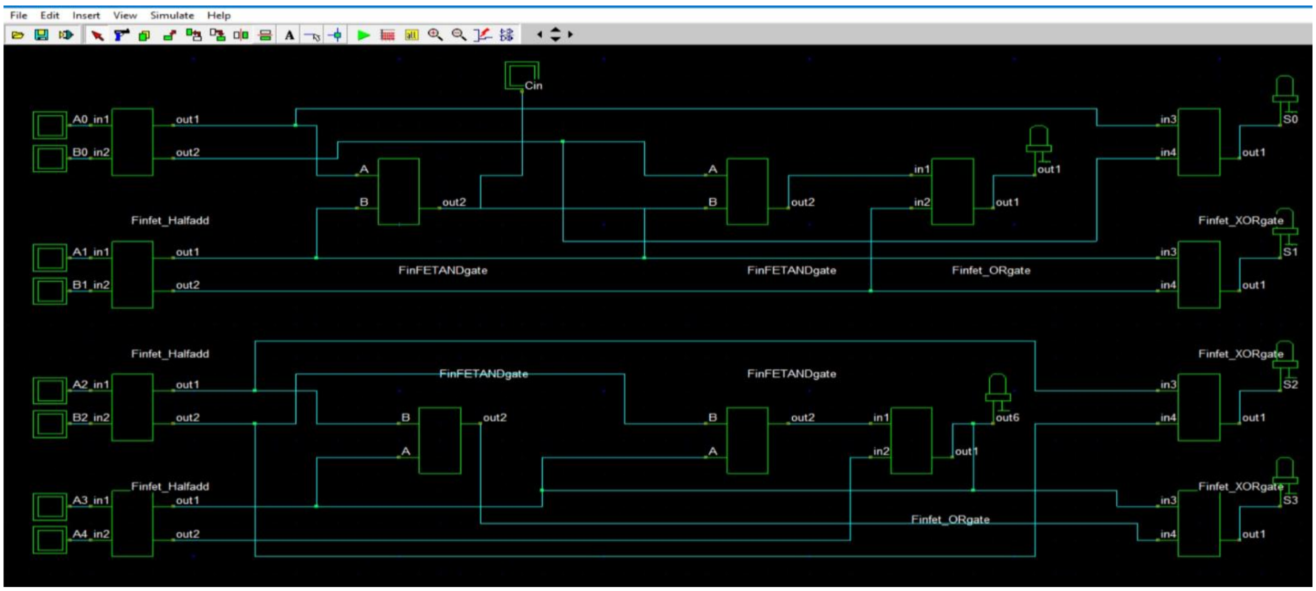


Fig.29: Ladner-Fischer Adder design using FinFET

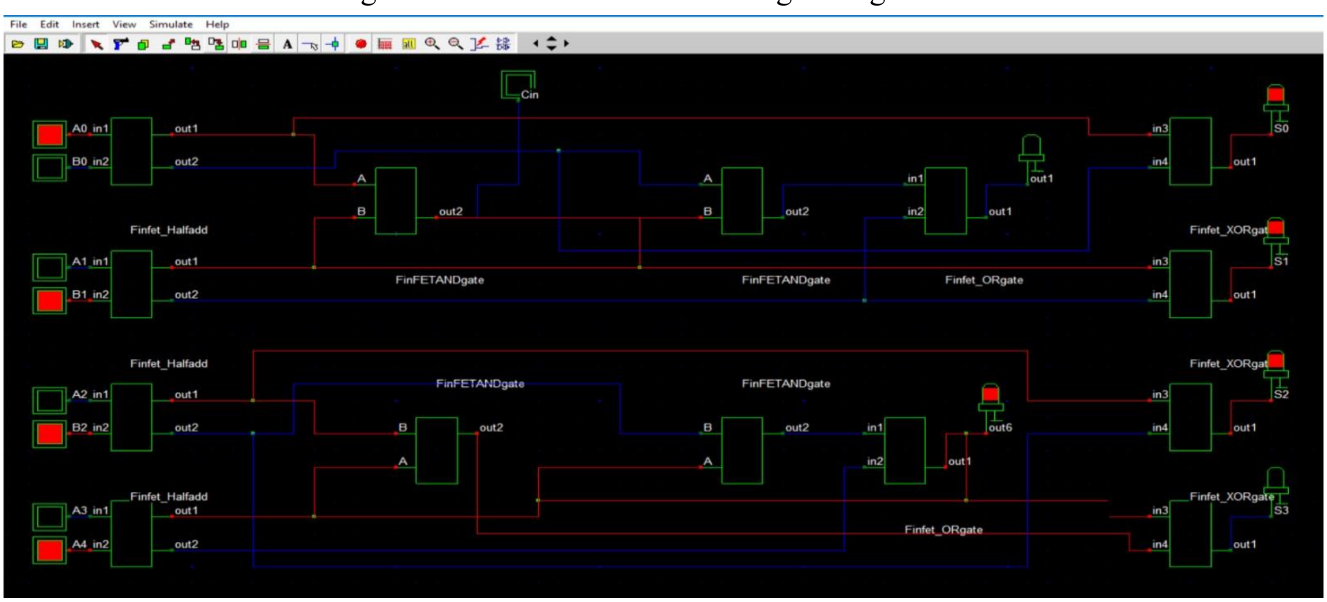


Fig.30: Execution of Ladner-Fischer Adder design using FinFET

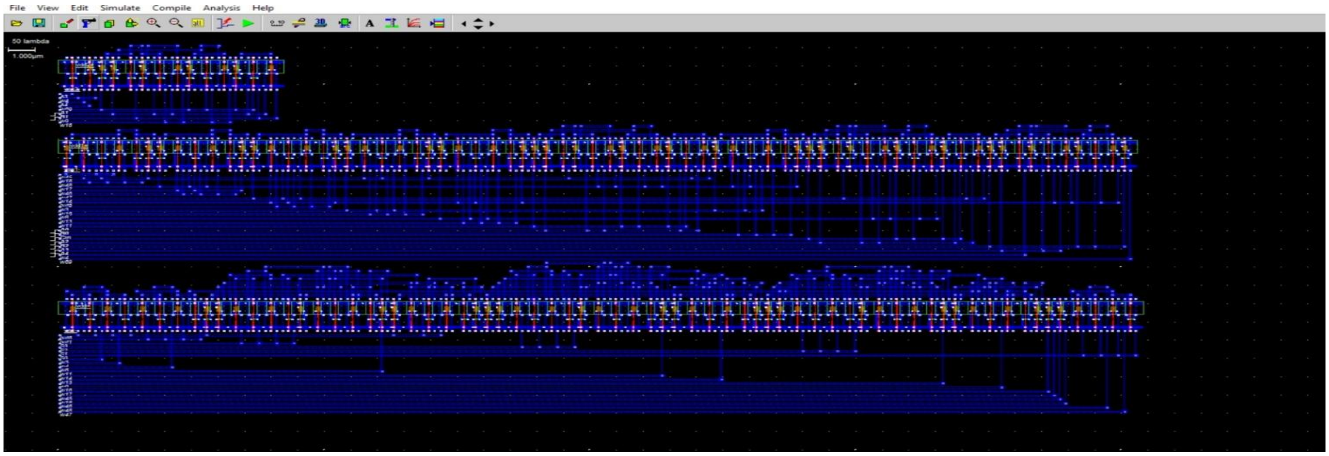


Fig.31: Layout Simulation of Ladner-Fischer Adder design using FinFET

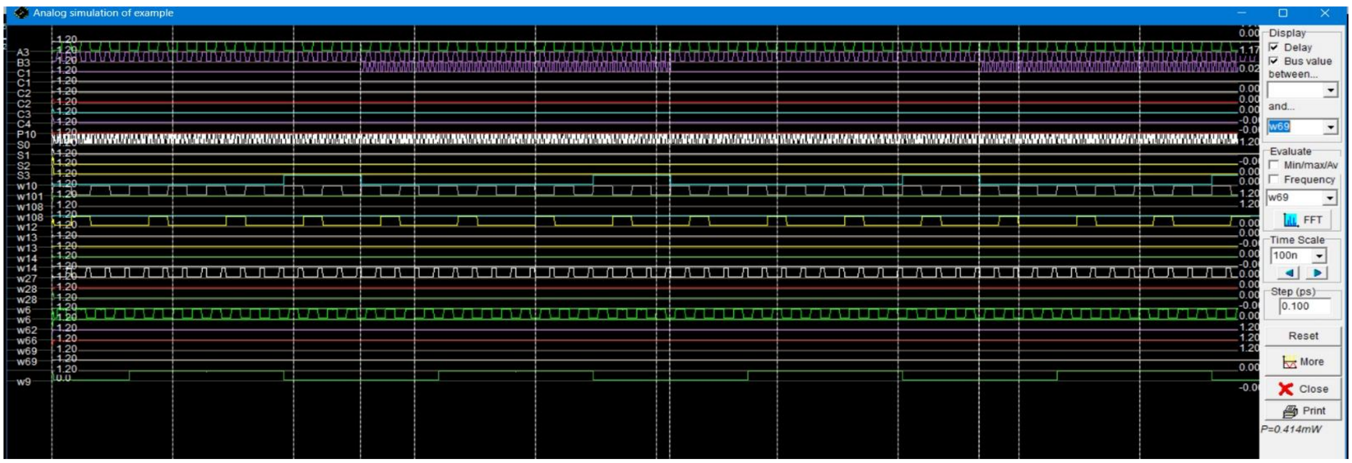


Fig.32: Power Results of Ladner-Fischer Adder using FinFET

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